

Problem 1: Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear operator which has an upper triangular matrix with respect to the basis $\beta = ((1,0,0), (1,1,1), (1,1,2))$. Find an orthonormal basis of $\mathbb{R}^3$ with respect to which T has an upper triangular.

Digression

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Let V be a finite dimensional inner product space and $\beta = (v_1, \dots, v_n)$ s.t
given $T \in \mathcal{L}(V)$, $[T]_{\beta}^{\beta}$ is upper triangular.

$$\Rightarrow T v_k \in \text{span}(v_1, \dots, v_k)$$

Apply Gram - Schmidt Orthonormalization to β and
obtain $\beta' = (w_1, \dots, w_n)$.

Then $\text{span}(w_1, \dots, w_k) = \text{span}(v_1, \dots, v_k) \quad \forall 1 \leq k \leq n.$

We are interested in Tw_k

$$w_k \in \text{span}(v_1, \dots, v_k)$$

$$w_k = a_1 v_1 + \dots + a_k v_k$$

$$Tw_k = a_1 Tv_1 + \dots + a_k Tv_k.$$

$$\begin{aligned} T v_j \in \text{span}\{v_1, \dots, v_j\} &\subseteq \text{span}\{v_1, \dots, v_k\} \\ &= \text{span}\{w_1, \dots, w_k\} \\ &\quad \forall 1 \leq j \leq k \end{aligned}$$

$$\Rightarrow Tw_k \in \text{span}(w_1, \dots, w_k)$$

$$\Rightarrow [T]_{\beta}^{\beta'} \text{ is upper triangular}$$

Solution to Problem 1: $T \in \mathcal{L}(\mathbb{R}^3)$ upper triangular

w.r.t $\beta = \left(\underset{v_1}{(1, 0, 0)}, \underset{v_2}{(1, 1, 1)}, \underset{v_3}{(1, 1, 2)} \right)$

$$w_1' = v_1$$

$$w'_2 = (1, 1, 1) - \frac{\langle (1, 1, 1), (1, 0, 0) \rangle}{1} \cdot (1, 0, 0)$$

$$= (1, 1, 1) - (1, 0, 0) = (0, 1, 1)$$

$$w'_3 = (1, 1, 2) - \langle (1, 1, 2), (1, 0, 0) \rangle (1, 0, 0) \\ - \frac{\langle (1, 1, 2), (0, 1, 1) \rangle}{2} (0, 1, 1)$$

$$= (1, 1, 2) - (1, 0, 0) - \frac{3}{2} (0, 1, 1)$$

$$= (0, 1, 2) - (0, \frac{3}{2}, \frac{3}{2})$$

$$= (0, -\frac{1}{2}, \frac{1}{2}) = \frac{1}{2} (0, -1, 1)$$

$$w_1 = (1, 0, 0)$$

$$w_2 = \frac{1}{\sqrt{2}} (0, 1, 1)$$

$$w_3 = \frac{1}{\sqrt{2}} (0, -1, 1)$$

T will be upper triangular w.r.t the orthonormal basis
 $\beta' = (w_1, w_2, w_3)$. $\square$

Exercise: Let V be an finite dimensional inner product space and let β' be obtained by the Gram Schmidt orthonormalization of a basis β .
The what can we say about $[I]_{\beta}^{\beta'}$

Problem 2: Let $W = \text{Span}\{(1, 0, i), (1, 2, 1)\}$ in $\mathbb{C}^3$.
Then compute an orthonormal basis of $W^\perp$.

Solution: $\beta = \left\{ \overset{v_1}{(1, 0, i)}, \overset{v_2}{(1, 2, 1)}, \overset{v_3}{(0, 0, 1)} \right\}$ be
a basis obtained by extending the basis of W .

Let $\beta' = (w_1, w_2, w_3)$ be obtained by
the G-S orthonormalization. Then

$\{w_3\}$ will be a basis of $w^\perp$

$$\begin{aligned} \text{(Notice that } \dim(w^\perp) &= \dim(\mathbb{C}^3) - \dim(w) \\ &= 3 - 2 = 1 \end{aligned}$$

$$\beta = \left((1, 0, i), (1, 2, 1), (0, 0, 1) \right)$$

$$w'_1 = (1, 0, i)$$

$$\begin{aligned} w'_2 &= (1, 2, 1) - \frac{\langle (1, 2, 1), (1, 0, i) \rangle}{2} (1, 0, i) \\ &= (1, 2, 1) - \frac{1-i}{2} (1, 0, i) \end{aligned}$$

$$= \left(\frac{1+i}{2}, 2, \frac{1-i}{2} \right) = \frac{1}{2} (1+i, 4, 1-i)$$

$\frac{1}{2} + 4 + \frac{1}{2} = 5$

$$w'_3 = (0, 0, 1) - \frac{\langle (0, 0, 1), (1, 0, i) \rangle}{2} (1, 0, i)$$

$$- \frac{\langle (0, 0, 1), (1+i, 4, 1-i) \rangle}{5 \times 2} \left(\frac{1+i}{2}, 2, \frac{1-i}{2} \right)$$

$$= (0, 0, 1) + \frac{i}{2} (1, 0, i) - \frac{i+1}{10} \left(\frac{1+i}{2}, 2, \frac{1-i}{2} \right)$$

$$= (i/2, 0, 1/2) - \quad "$$

$$= \frac{i}{2} (1, 0, -i) - \left(\frac{i}{10}, \frac{i+1}{5}, \frac{1}{10} \right)$$

$$= \left(\frac{2i}{5}, -\frac{(i+1)}{5}, \frac{2}{5} \right)$$

$$\left(i, -\frac{(1+i)}{2}, 1 \right)$$

$$\frac{\left(1 + \frac{1}{2} + 1 \right)^{1/2}}{}$$

$$= \frac{\sqrt{2}}{\sqrt{5}} \left(i, -\frac{(1+i)}{2}, 1 \right)$$

— ■ .

Problem 3: In $\mathbb{R}^3$, compute the distance of the vector
 $u = (2, 1, 3)$ from the subspace
 $W = \{ (x, y, z) : x + 3y - 2z = 0 \}.$

Solution: Let $\beta = ((-3, 1, 0), (2, 0, 1))$ be a
basis of W .

Let $\beta' = (w_1, w_2)$ be obtained by G-S-O.

$$\begin{aligned} w'_2 &= (2, 0, 1) - \frac{\langle (2, 0, 1), (-3, 1, 0) \rangle}{10} (-3, 1, 0) \\ &= (2, 0, 1) + \frac{3}{5} (-3, 1, 0) \\ &= \left(\frac{1}{5}, \frac{3}{5}, 1 \right) \end{aligned}$$

Then the projection of $(2, 1, 3)$

$$\text{Let } v = \langle (2, 1, 3), w_1 \rangle w_1 + \langle (2, 1, 3), w_2 \rangle w_2$$

$$\frac{1}{10} \langle (2, 1, 3), (-3, 1, 0) \rangle (-3, 1, 0) +$$

$$\frac{\langle (2, 1, 3), (1, 3, 5) \rangle}{35} (1, 3, 5)$$

$$= -\frac{1}{2} (-3, 1, 0) + \frac{4}{7} (1, 3, 5)$$

$$= \left(\frac{3}{2} + \frac{4}{7}, \frac{12}{7} - \frac{1}{2}, \frac{20}{7} \right)$$

$$= \left(\frac{29}{14}, \frac{17}{14}, \frac{20}{7} \right)$$

Hence distance of $(2, 1, 3)$ to W is given by

$$\left\| (2, 1, 3) - \left(\frac{29}{14}, \frac{17}{14}, \frac{20}{7} \right) \right\|$$

$$= \left\| \left(-\frac{1}{14}, -\frac{3}{14}, \frac{1}{7} \right) \right\|$$

$$= \left(\frac{1}{14^2} (1 + 9 + 4) \right)^{1/2}$$

=

$$\underline{\underline{\frac{1}{\sqrt{14}}}}$$

Problem 4: Assume that $(v_1, \dots, v_k)$ is an orthonormal set in an inner product space V .

Let $u \in V$. Then prove that

$$\sum_{j=1}^k |\langle u, v_j \rangle|^2 \leq \|u\|^2$$

with equality iff $u \in \text{span}(v_1, \dots, v_k)$.

Proof: Let $v = \langle u, v_1 \rangle v_1 + \dots + \langle u, v_k \rangle v_k$.

Let $w = u - v$

$$\langle w, v_j \rangle = \langle u - v, v_j \rangle = \langle u, v_j \rangle - \langle u, v_j \rangle = 0$$

$$\Rightarrow u - v \perp v_j \quad \forall j = 1, \dots, k.$$

If $W = \text{span}(v_1, \dots, v_k)$

then $(u - v) \perp W$.

In particular $w \perp v$

Hence

$$u = v + w$$

where

$$v \perp w$$

$$\|u\|^2 = \|v\|^2 + \|w\|^2$$

$$\Rightarrow \|u\|^2 \geq \|v\|^2$$

Recall

$$v = \langle u, v_1 \rangle v_1 + \dots + \langle u, v_k \rangle v_k$$

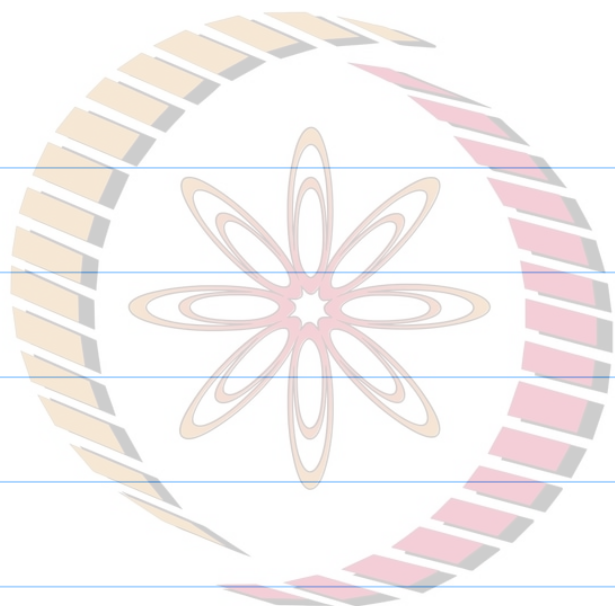
$$\|v\|^2 = \sum_{j=1}^k |\langle u, v_j \rangle|^2$$

with equality iff $w = 0$

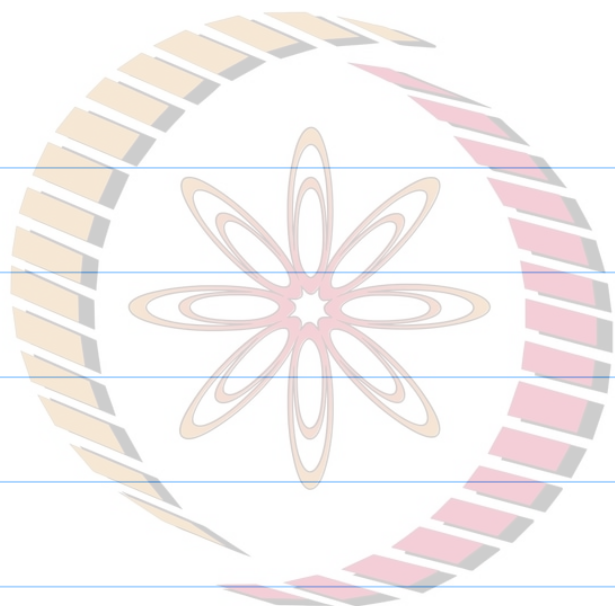
$$\Leftrightarrow u = v$$

$$\Leftrightarrow u \in W, \quad \text{—} \quad \blacksquare.$$

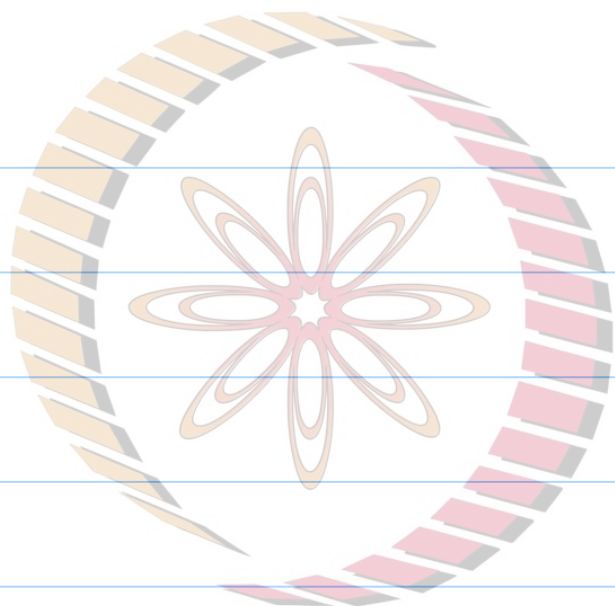
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